PROJECT BASED EXPERIMENTAL LEARNING

COURSE- DATABASE MANAGEMENT SYSTEM (NoSQL / MongoDB)

PROJECT

TOPIC- MONGODB STUDENT DATABASE OPERATIONS

IBM

SUBMITTED TO- MR. Harendra Singh Rajpoot
Submitted by- Aarsh
BCADS27
Roll no. - 1250258003
3rd semester
Babu Banarasi Das University
Lucknow, U.P

Code:

```
db.students.insertOne({
rollNo: 101,
name: "Vishnu",
age: 22,
department: "CSE", marks: 85 })
```

```
{
  acknowledged: true,
  insertedId: ObjectId('6abb3e882a6ea1af20ca00d3')
}
```

2. Insert Many

```
db.students.insertMany([ { rollNo: 102, name: "Saraswati", age:
21, department: "IT", marks: 78 },
{ rollNo: 103, name: "Govind", age: 20, department: "CSE", marks: 92 },
{ rollNo: 104, name: "Parvati", age: 19, department: "ECE", marks: 81 },
{ rollNo: 105, name: "Hari", age: 22, department: "ME", marks: 74 },
{ rollNo: 106, name: "Radha", age: 21, department: "CSE", marks: 88 },
{ rollNo: 107, name: "Mohan", age: 20, department: "IT", marks: 69 },
{ rollNo: 108, name: "Kamala", age: 19, department: "ECE", marks: 95 },
{ rollNo: 109, name: "Shankar", age: 22, department: "CSE", marks: 83 },
{ rollNo: 110, name: "Yamuna", age: 21, department: "IT", marks: 76 },
{ rollNo: 111, name: "Krishna", age: 20, department: "ME", marks: 72 },
{ rollNo: 112, name: "Sita", age: 19, department: "CSE", marks: 89 },
{ rollNo: 113, name: "Rajendra", age: 22, department: "ECE", marks: 67 },
{ rollNo: 114, name: "Kaveri", age: 21, department: "IT", marks: 91 },
{ rollNo: 115, name: "Gopal", age: 20, department: "CSE", marks: 79 },
{ rollNo: 116, name: "Savitri", age: 19, department: "ECE", marks: 84 },
{ rollNo: 117, name: "Bhairav", age: 22, department: "ME", marks: 73 },
{ rollNo: 118, name: "Rukmini", age: 21, department: "CSE", marks: 96 },
{ rollNo: 119, name: "Indra", age: 20, department: "IT", marks: 68 },
{ rollNo: 120, name: "Chandra", age: 19, department: "ECE", marks: 87 },
{ rollNo: 121, name: "Manohar", age: 22, department: "CSE", marks: 75 },
{ rollNo: 122, name: "Gauri", age: 21, department: "IT", marks: 90 },
{ rollNo: 123, name: "Vasudev", age: 20, department: "ME", marks: 71 },
{ rollNo: 124, name: "Uma", age: 19, department: "CSE", marks: 86 },
{ rollNo: 125, name: "Dwarka", age: 22, department: "ECE", marks: 80 },
{ rollNo: 126, name: "Kunti", age: 21, department: "IT", marks: 94 },
{ rollNo: 127, name: "Bhupendra", age: 20, department: "CSE", marks: 77 },
{ rollNo: 128, name: "Draupadi", age: 19, department: "ECE", marks: 82 },
{ rollNo: 129, name: "Ramesh", age: 22, department: "ME", marks: 65 },
{ rollNo: 130, name: "Lakshmi", age: 21, department: "CSE", marks: 93 },
{ rollNo: 131, name: "Suresh", age: 20, department: "IT", marks: 70 },
{ rollNo: 132, name: "Kausalya", age: 19, department: "ECE", marks: 89 },
{ rollNo: 133, name: "Prithvi", age: 22, department: "CSE", marks: 81 },
{ rollNo: 134, name: "Sarala", age: 21, department: "IT", marks: 85 },
{ rollNo: 135, name: "Mahesh", age: 20, department: "ME", marks: 78 },
{ rollNo: 136, name: "Janaki", age: 19, department: "CSE", marks: 97 },
{ rollNo: 137, name: "Narayan", age: 22, department: "ECE", marks: 74 },
{ rollNo: 138, name: "Urmila", age: 21, department: "IT", marks: 88 },
{ rollNo: 139, name: "Devendra", age: 20, department: "CSE", marks: 66 },
{ rollNo: 140, name: "Mandakini", age: 19, department: "ECE", marks: 92 },
{ rollNo: 141, name: "Ishwar", age: 22, department: "ME", marks: 69 },
{ rollNo: 142, name: "Shakuntala", age: 21, department: "CSE", marks: 84 },
{ rollNo: 143, name: "Trilok", age: 20, department: "IT", marks: 79 },
{ rollNo: 144, name: "Kamalini", age: 19, department: "ECE", marks: 91 },
{ rollNo: 145, name: "Ranjit", age: 22, department: "CSE", marks: 87 },
{ rollNo: 146, name: "Vidyawati", age: 21, department: "IT", marks: 73 },
{ rollNo: 147, name: "Chaturbhuj", age: 20, department: "ECE", marks: 86 },
{ rollNo: 148, name: "Sushila", age: 19, department: "ME", marks: 68 },
{ rollNo: 149, name: "Omkar", age: 22, department: "CSE", marks: 90 },
{ rollNo: 150, name: "Tulsi", age: 21, department: "IT", marks: 76 },
{ rollNo: 151, name: "Balram", age: 20, department: "ECE", marks: 83 }])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6abb3edc2a6ea1af20ca00d4'),
    '1': ObjectId('6abb3edc2a6ea1af20ca00d5'),
    '2': ObjectId('6abb3edc2a6ea1af20ca00d6'),
    '3': ObjectId('6abb3edc2a6ea1af20ca00d7'),
    '4': ObjectId('6abb3edc2a6ea1af20ca00d8'),
    '5': ObjectId('6abb3edc2a6ea1af20ca00d9'),
    '6': ObjectId('6abb3edc2a6ea1af20ca00da'),
    '7': ObjectId('6abb3edc2a6ea1af20ca00db'),
    '8': ObjectId('6abb3edc2a6ea1af20ca00dc'),
    '9': ObjectId('6abb3edc2a6ea1af20ca00dd'),
    '10': ObjectId('6abb3edc2a6ea1af20ca00de'),
    '11': ObjectId('6abb3edc2a6ea1af20ca00df'),
    '12': ObjectId('6abb3edc2a6ea1af20ca00e0'),
    '13': ObjectId('6abb3edc2a6ea1af20ca00e1'),
    '14': ObjectId('6abb3edc2a6ea1af20ca00e2'),
    '15': ObjectId('6abb3edc2a6ea1af20ca00e3'),
    '16': ObjectId('6abb3edc2a6ea1af20ca00e4'),
    '17': ObjectId('6abb3edc2a6ea1af20ca00e5'),
    '18': ObjectId('6abb3edc2a6ea1af20ca00e6'),
    '19': ObjectId('6abb3edc2a6ea1af20ca00e7'),
    '20': ObjectId('6abb3edc2a6ea1af20ca00e8'),
    '21': ObjectId('6abb3edc2a6ea1af20ca00e9'),
    '22': ObjectId('6abb3edc2a6ea1af20ca00ea'),
```

3. Update One Code:

```
db.students.updateOne( {
rollNo: 101 },
{ $set: { marks: 90 } }
)
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

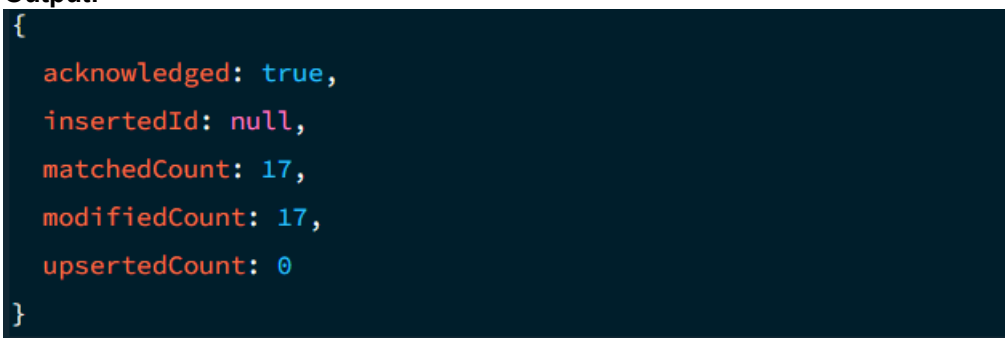
Output:

4. Update Many

Code:

```
db.students.updateMany( { department: "CSE"
},
{ $set: { status: "Active" } }
)
```

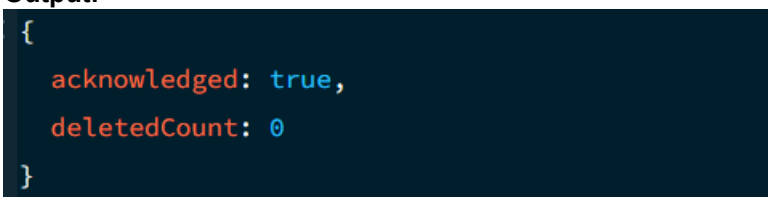
Output:



5. Delete One

Code: `db.students.deleteOne({ rollNo: 151 })`

Output:



6. Delete Many

.

Code: `db.students.deleteMany({ marks: { $lt: 40 } })`

Output:

```
{
  _id: ObjectId('6ab5fc9f00a9d2b17507e62f'),
  rollNo: 5,
  name: 'Neha',
  age: 21,
  department: 'ECE',
  marks: 88
}
{
  _id: ObjectId('6ab5fc9f00a9d2b17507e632'),
  rollNo: 8,
  name: 'Vikas',
  age: 21,
  department: 'ECE',
  marks: 90
}
{
  _id: ObjectId('6ab5fc9f00a9d2b17507e634'),
  rollNo: 10,
  name: 'Karan',
  age: 23,
  department: 'IT',
  marks: 83
}
```

7. Query Operations

Marks greater than 80:

```
db.students.find({ marks: { $gt:
80 } })
```

Output:



Marks less than 80:

```
db.students.find({ marks: { $lt:
80 } })
```

Output:

```
{
  _id: ObjectId('6ab5fc9f00a9d2b17507e64a'),
  rollNo: 32,
  name: 'Manish',
  age: 22,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('6abb3edc2a6ea1af20ca00f4'),
  rollNo: 134,
  name: 'Sarala',
  age: 21,
  department: 'IT',
  marks: 85
}
db.students.find({ marks: { $gte: 80 } })
{
  _id: ObjectId('6ab3a61c2545dcfec9ce2e54'),
  id: 23,
  name: 'Varun',
  age: 21,
  department: 'ECE',
  marks: 80
}
```

Marks equal to 85:

```
db.students.find({ marks: { $eq:
85 } })
```

Output:

```
{
  _id: ObjectId('6ab3a61c2545dcfec9ce2e54'),
  id: 23,
  name: 'Varun',
  age: 21,
  department: 'ECE',
  marks: 80
}
{
  _id: ObjectId('6ab3a61c2545dcfec9ce2e6e'),
  id: 49,
  name: 'Varun',
  age: 20,
  department: 'IT',
  marks: 80
}
{
  _id: ObjectId('6ab557107f96d09a2a0f3fd1'),
  rollNo: 22,
  name: 'Sahil',
  age: 21,
  department: 'IT',
  marks: 80
}
```

Marks greater than or equal to 80:

```
db.students.find({ marks: { $gte: 80 } })
```

Output:

```
{
  _id: ObjectId('6aaa68b92a03d67cd4ab5dff'),
  name: 'Rahul',
  age: 22,
  marks: 72,
  department: 'ECE',
  skills: [
    'Python',
    'SQL'
  ]
}
{
  _id: ObjectId('6aaa68b92a03d67cd4ab5e01'),
  name: 'Neha',
  age: 23,
  marks: 65,
  department: 'ME',
  skills: [
    'AutoCAD',
    'Python'
  ]
}
```


Marks greater than 80 AND age less than 23:

```
db.students.find({ $and: [ { marks: { $gt: 80 } },
{ age: { $lt: 23 } }
]
})
```

Output:

```
{
  _id: ObjectId('6aaa68b92a03d67cd4ab5dff'),
  name: 'Rahul',
  age: 22,
  marks: 72,
  department: 'ECE',
  skills: [
    'Python',
    'SQL'
  ]
}
{
  _id: ObjectId('6ab3a61c2545dcfec9ce2e3f'),
  id: 2,
  name: 'Vivaan',
  age: 21,
  department: 'ECE',
  marks: 78
}
{
  _id: ObjectId('6ab3a61c2545dcfec9ce2e41'),
  id: 4,
  name: 'Arjun',
  age: 22,
  department: 'ME',
  marks: 74
}
```


Department is CSE OR ECE:

```
db.students.find({ $or: [ { department: "CSE" },
{ department: "ECE" }
]
})
```

Output:

CSE and ECE department students are returned.

```
{
  _id: ObjectId('6aaa68b92a03d67cd4ab5dff'),
  name: 'Rahul',
  age: 22,
  marks: 72,
  department: 'ECE',
  skills: [
    'Python',
    'SQL'
  ]
}
{
  _id: ObjectId('6ab3a61c2545dcfec9ce2e3f'),
  id: 2,
  name: 'Vivaan',
  age: 21,
  department: 'ECE',
  marks: 78
}
{
  _id: ObjectId('6ab3a61c2545dcfec9ce2e47'),
  id: 10,
  name: 'Neha',
  age: 22,
  department: 'ECE',
  marks: 76
}
```

Marks NOT greater than 80:

```
db.students.find({ marks: { $not: { $gt:
80 } } })
```

Output:

```
{
  _id: ObjectId('6aaa68b92a03d67cd4ab5dff'),
  name: 'Rahul',
  age: 22,
  marks: 72,
  department: 'ECE',
  skills: [
    'Python',
    'SQL'
  ]
}
{
  _id: ObjectId('6aaa68b92a03d67cd4ab5e01'),
  name: 'Neha',
  age: 23,
  marks: 65,
  department: 'ME',
  skills: [
    'AutoCAD',
    'Python'
  ]
}
```